

Database Project Proposal Form

Project #		
Project Title:	Student Information System	
TA Name:	Mai Sherif – Maryam Ashraf	
Students' names:	<i>ID Number</i>	<i>Name</i>
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2-		
3-		
4-		
5-		
Statement of Problem:	University academic processes are often managed using fragmented or manual systems, leading to data redundancy, inconsistencies, and difficulties in tracking students' academic activities such as enrollment, attendance, exams, and grades. There is a need for a centralized, secure, and efficient database system to manage all academic operations in an organized way with proper access control.	
Brief Description of the Project:	This project is a Student Information System (SIS) database designed to manage university academic operations. It includes core modules such as students, instructors, courses, registration, attendance, exams, and grades. The system is built using relational database principles with normalization, constraints, views, indexing, and role-based security to ensure data integrity, performance, and secure access for different user roles.	
Objectives:	- Design a normalized relational database for university management. - Store and manage students, instructors, courses, and academic records efficiently. - Track student enrollment, attendance, exams, and grades. - Improve query performance using indexes. - Provide simplified data access using views. - Implement role-based security to control data access for Admin, Instructor, TA, and Student. - Ensure data integrity using constraints and relationships.	
Technical Approach:		

	• Store data in SQL Server • ER Diagram
Expected Deliverables:	Entity-Relationship (ER) model of the system. Seed data for testing and demonstration.